

Which of the following is not an aggregate operation?

☐ SUM

☐ PROD

☐ MAX

☐ AVG

Consider the given relation r

1 point

A	B	C	D
1	1	3	4
3	4	5	1
1	1	2	3
3	2	5	1

Figure 1: Relation r

What will be the output of the following relational operation on r?

$\sigma_{(A=B \vee C > 2)}(r)$

☐

A	B	C	D
1	1	3	4
1	1	2	3

☐

A	B	C	D
3	4	5	1
3	2	5	1

☐

A	B	C	D

☐

A	B	C	D
3	4	5	1
3	2	5	1
1	1	3	4

Consider the given relation r

1 point

A	B	C	D
1	1	3	4
3	4	5	1
1	1	2	3
3	2	5	1

Figure 2: Relation r

How many rows will be there in the output of the following relational operation on r?

$\pi_{A,D}(r)$

☐ 4

☐ 0

☐ 3

☐ 2

Consider the two relations r and s.

1 point

A	B	C	D
0	9	0	6
2	2	2	0
1	1	1	1

A	B	C
1	1	3
3	4	5
1	1	2
3	2	5

Figure 3: Relation r, Relation s

How many columns will be there in the union of the above two relations?

☐ 4

- ☐ 3
- ☐ 7
- ☐ Union not possible

Consider the two relations r and s.

1 point

A	B	C
0	9	0
2	2	2
1	1	1

A	B	C
0	Cat	0
2	Dog	2
1	Lion	1

Figure 4: Relation r , Relation s

In relation r all three attributes have integer domain.

In relation s attribute A, C has integer domain and attribute B has the type varchar as domain.

How many columns will be there in the union of the above two relations?

- ☐ 4
- ☐ 3
- ☐ 7
- ☐ Union not possible

Consider the two relations r and s.

1 point

A	B	C
0	9	0
2	2	2
1	1	1

A	B	C
0	Cat	0
2	Dog	2
1	Lion	1

Figure 5: Relation r, Relation s

How many columns and rows will be there in the Cartesian product of the above two relations?

- ☐ Columns = 6, Rows = 9
- ☐ Columns = 3, Rows = 9
- ☐ Columns = 3, Rows = 6
- ☐ Columns = 6, Rows = 6

If two relations are not UNION-compatible then which of the following operation(s) can not be performed? **1 point**

- ☐ UNION
- ☐ CARTESIAN PRODUCT
- ☐ SET DIFFERENCE
- ☐ INTERSECTION

Consider two relations **r** and **s** as shown in Figure 6.

A	B	C
a1	b1	c1
a2	b2	c1
a3	b1	c3
a1	b3	c2
a2	b4	c2
a4	b1	a3

X	Y	A	C
x1	y1	a1	c1
x2	y2	a1	c2
x3	y3	a2	c2
x3	y3	a3	c2
x1	y4	a3	c3

Figure 6: Relation **r** and **s**

How many rows will be selected by relational operation $r \bowtie s$

1 point

Consider two relations $s(A, B, C)$ and $r(P, Q, C)$ such that relation s has 30 rows and relation r has 12 rows. What is the maximum number of rows that are possible in $r \bowtie s$?

1 point

Consider two relations $s(A, B, C)$ and $r(P, Q, C)$ such that relation s has 30 rows and relation r has 12 rows. What is the minimum number of rows that are possible in $r \bowtie s$?

1 point